

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 15A

CHIRALITY, OPTICAL ACTIVITY & REACTION STEREOCHEMISTRY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15A: Chirality, Optical Activity & Reaction Stereochemistry
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	201 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Chiral Centre Identification****[3 Marks]****WCH14/01/Jan23/Q15**

Lactic acid, 2-hydroxypropanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, is found in sour milk.

- (a) Draw the structural formula of 2-hydroxypropanoic acid and circle the chiral carbon atom. [1]

- (b) Draw 3D wedge-and-dash diagrams for the two enantiomers of 2-hydroxypropanoic acid. [2]

Q2. 2. Polarimetry and Racemic Mixtures**[4 Marks]****WCH14/01/Oct22/Q16**

A sample of 2-chlorobutane was synthesized in a laboratory.

- (a) Describe how a polarimeter can be used to show that a sample of 2-chlorobutane is optically active. [2]

- (b) Explain why a sample of 2-chlorobutane prepared by reacting but-2-ene with HCl is optically inactive. [2]

Q3. 3. Chirality & Enantiomer Property Step 3**[3 Marks]****WCH14/01/Sample/Q3**

Consider organic molecule 3 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q4. 4. Chirality & Enantiomer Property Step 4**[3 Marks]****WCH14/01/Sample/Q4**

Consider organic molecule 4 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q5. 5. Chirality & Enantiomer Property Step 5**[3 Marks]****WCH14/01/Sample/Q5**

Consider organic molecule 5 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q6. 6. Chirality & Enantiomer Property Step 6**[3 Marks]****WCH14/01/Sample/Q6**

Consider organic molecule 6 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q7. 7. Chirality & Enantiomer Property Step 7**[3 Marks]****WCH14/01/Sample/Q7**

Consider organic molecule 7 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q8. 8. Chirality & Enantiomer Property Step 8**[3 Marks]****WCH14/01/Sample/Q8**

Consider organic molecule 8 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q9. 9. Chirality & Enantiomer Property Step 9**[3 Marks]**
WCH14/01/Sample/Q9

Consider organic molecule 9 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q10. 10. Chirality & Enantiomer Property Step 10**[3 Marks]**
WCH14/01/Sample/Q10

Consider organic molecule 10 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q11. 11. Chirality & Enantiomer Property Step 11**[3 Marks]**
WCH14/01/Sample/Q11

Consider organic molecule 11 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q12. 12. Chirality & Enantiomer Property Step 12**[3 Marks]**
WCH14/01/Sample/Q12

Consider organic molecule 12 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q13. 13. Chirality & Enantiomer Property Step 13**[3 Marks]**
WCH14/01/Sample/Q13

Consider organic molecule 13 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q14. 14. Chirality & Enantiomer Property Step 14

[3 Marks]

WCH14/01/sample/Q14

Consider organic molecule 14 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q15. 15. Chirality & Enantiomer Property Step 15

[3 Marks]

WCH14/01/sample/Q15

Consider organic molecule 15 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q16. 16. Chirality & Enantiomer Property Step 16

[3 Marks]

WCH14/01/sample/Q16

Consider organic molecule 16 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

- (b) State one property in which the two enantiomers differ. [1]

Q17. 17. Chirality & Enantiomer Property Step 17

[3 Marks]

WCH14/01/sample/Q17

Consider organic molecule 17 with a single chiral centre.

- (a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q18. 18. Chirality & Enantiomer Property Step 18

[3 Marks]

WCH14/01/sample/Q18

Consider organic molecule 18 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q19. 19. Chirality & Enantiomer Property Step 19

[3 Marks]

WCH14/01/sample/Q19

Consider organic molecule 19 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q20. 20. Chirality & Enantiomer Property Step 20

[3 Marks]

WCH14/01/sample/Q20

Consider organic molecule 20 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q21. 21. Chirality & Enantiomer Property Step 21

[3 Marks]

WCH14/01/sample/Q21

Consider organic molecule 21 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q22. 22. Chirality & Enantiomer Property Step 22

[3 Marks]

WCH14/01/sample/Q22

Consider organic molecule 22 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q23. 23. Chirality & Enantiomer Property Step 23

[3 Marks]

WCH14/01/sample/Q23

Consider organic molecule 23 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q24. 24. Chirality & Enantiomer Property Step 24

[3 Marks]

WCH14/01/sample/Q24

Consider organic molecule 24 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

Q25. 25. Chirality & Enantiomer Property Step 25

[3 Marks]

WCH14/01/sample/Q25

Consider organic molecule 25 with a single chiral centre.

(a) State two physical properties that are identical for both enantiomers. [2]

(b) State one property in which the two enantiomers differ. [1]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: SN1 vs SN2 Stereochemical Outcomes****[5 Marks]****26****WCH14/01/Hard/Q26**

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q27. 27. A* Challenge: SN1 vs SN2 Stereochemical Outcomes**[5 Marks]****27****WCH14/01/Hard/Q27**

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q28. 28. A* Challenge: SN1 vs SN2 Stereochemical Outcomes**[5 Marks]****28****WCH14/01/Hard/Q28**

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q29. 29. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
29**[5 Marks]**
WCH14/01/Hard/Q29

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

- (a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

- (b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q30. 30. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
30**[5 Marks]**
WCH14/01/Hard/Q30

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

- (a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

- (b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q31. 31. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 31**[5 Marks]**
WCH14/01/Hard/Q31

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

- (a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

- (b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q32. 32. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
32**[5 Marks]**
WCH14/01/Hard/Q32

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q33. 33. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
33

[5 Marks]
WCH14/01/Hard/Q33

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q34. 34. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
34

[5 Marks]
WCH14/01/Hard/Q34

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q35. 35. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
35

[5 Marks]
WCH14/01/Hard/Q35

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q36. 36. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
36

[5 Marks]
WCH14/01/Hard/Q36

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q37. 37. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
37

[5 Marks]
WCH14/01/Hard/Q37

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q38. 38. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
38

[5 Marks]
WCH14/01/Hard/Q38

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q39. 39. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
39

[5 Marks]
WCH14/01/Hard/Q39

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q40. 40. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
40

[5 Marks]
WCH14/01/Hard/Q40

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q41. 41. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 41

[5 Marks]
WCH14/01/Hard/Q41

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q42. 42. A* Challenge: SN1 vs SN2 Stereochemical Outcomes**42****[5 Marks]****WCH14/01/Hard/Q42**

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q43. 43. A* Challenge: SN1 vs SN2 Stereochemical Outcomes**43****[5 Marks]****WCH14/01/Hard/Q43**

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q44. 44. A* Challenge: SN1 vs SN2 Stereochemical Outcomes**44****[5 Marks]****WCH14/01/Hard/Q44**

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q45. 45. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
45**[5 Marks]**
WCH14/01/Hard/Q45

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q46. 46. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
46**[5 Marks]**
WCH14/01/Hard/Q46

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q47. 47. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
47**[5 Marks]**
WCH14/01/Hard/Q47

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q48. 48. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
48**[5 Marks]**
WCH14/01/Hard/Q48

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q49. 49. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
49

[5 Marks]
WCH14/01/Hard/Q49

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

Q50. 50. A* Challenge: SN1 vs SN2 Stereochemical Outcomes
50

[5 Marks]
WCH14/01/Hard/Q50

Hydrolysis of optically active (R)-2-bromobutane with aqueous NaOH yields a mixture of products.

(a) Explain why an SN1 mechanism leads to a racemic mixture (loss of optical activity). [3]

(b) Explain why an SN2 mechanism leads to complete inversion of optical configuration (Walden inversion). [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 15A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Chiral Centre Identification 1. (a) C2 circled (attached to -H, -OH, -CH ₃ , -COOH) (1). 1. (b) Tetrahedral 3D drawings showing non-superimposable mirror images (2).	[3]
Q2	Q2: 2. Polarimetry and Racemic Mixtures 2. (a) Pass plane-polarised light through the sample (1); optically active sample rotates the plane of polarisation (1). 2. (b) Electrophilic addition involves a planar carbocation intermediate (1); H ⁺ / Cl ⁻ can attack from top or bottom of the plane with equal probability, forming a 50:50 racemic mixture (1).	[4]
Q3	Q3: 3. Chirality & Enantiomer Property Step 3 3. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 3. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q4	Q4: 4. Chirality & Enantiomer Property Step 4 4. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 4. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q5	Q5: 5. Chirality & Enantiomer Property Step 5 5. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 5. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q6	Q6: 6. Chirality & Enantiomer Property Step 6 6. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 6. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q7	Q7: 7. Chirality & Enantiomer Property Step 7 7. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 7. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q8	Q8: 8. Chirality & Enantiomer Property Step 8 8. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 8. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q9	Q9: 9. Chirality & Enantiomer Property Step 9 9. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 9. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q10	Q10: 10. Chirality & Enantiomer Property Step 10 10. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 10. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q11	Q11: 11. Chirality & Enantiomer Property Step 11 11. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 11. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q12	Q12: 12. Chirality & Enantiomer Property Step 12 12. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 12. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]

Q13	Q13: 13. Chirality & Enantiomer Property Step 13 13. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 13. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q14	Q14: 14. Chirality & Enantiomer Property Step 14 14. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 14. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q15	Q15: 15. Chirality & Enantiomer Property Step 15 15. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 15. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q16	Q16: 16. Chirality & Enantiomer Property Step 16 16. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 16. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q17	Q17: 17. Chirality & Enantiomer Property Step 17 17. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 17. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q18	Q18: 18. Chirality & Enantiomer Property Step 18 18. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 18. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q19	Q19: 19. Chirality & Enantiomer Property Step 19 19. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 19. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q20	Q20: 20. Chirality & Enantiomer Property Step 20 20. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 20. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q21	Q21: 21. Chirality & Enantiomer Property Step 21 21. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 21. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q22	Q22: 22. Chirality & Enantiomer Property Step 22 22. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 22. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q23	Q23: 23. Chirality & Enantiomer Property Step 23 23. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 23. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q24	Q24: 24. Chirality & Enantiomer Property Step 24 24. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 24. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]
Q25	Q25: 25. Chirality & Enantiomer Property Step 25 25. (a) Melting point, boiling point, density, solubility in achiral solvents (any 2) (2). 25. (b) Rotation of plane-polarised light (opposite directions) / biological reaction with chiral enzymes (1).	[3]

Q26	Q26: 26. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 26 26. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 26. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q27	Q27: 27. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 27 27. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 27. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q28	Q28: 28. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 28 28. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 28. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q29	Q29: 29. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 29 29. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 29. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q30	Q30: 30. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 30 30. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 30. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q31	Q31: 31. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 31 31. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 31. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q32	Q32: 32. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 32 32. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 32. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q33	Q33: 33. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 33 33. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 33. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q34	Q34: 34. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 34 34. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 34. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]

Q35	Q35: 35. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 35 35. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 35. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q36	Q36: 36. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 36 36. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 36. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q37	Q37: 37. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 37 37. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 37. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q38	Q38: 38. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 38 38. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 38. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q39	Q39: 39. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 39 39. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 39. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q40	Q40: 40. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 40 40. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 40. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q41	Q41: 41. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 41 41. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 41. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q42	Q42: 42. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 42 42. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 42. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q43	Q43: 43. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 43 43. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 43. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]

Q44	Q44: 44. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 44 44. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 44. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q45	Q45: 45. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 45 45. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 45. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q46	Q46: 46. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 46 46. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 46. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q47	Q47: 47. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 47 47. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 47. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q48	Q48: 48. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 48 48. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 48. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q49	Q49: 49. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 49 49. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 49. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]
Q50	Q50: 50. A* Challenge: SN1 vs SN2 Stereochemical Outcomes 50 50. (a) SN1 forms a trigonal planar carbocation intermediate (1). Nucleophile OH ⁻ attacks from top or bottom face with 50:50 probability (1) producing equal amounts of (R) and (S) enantiomers → racemate (1). 50. (b) SN2 is a single-step concerted reaction with backside attack of OH ⁻ opposite the Br leaving group (1), causing inversion of configuration (1).	[5]

SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 15A**FAQ #1: Identifying Chiral Carbons in Rings****Category: Ring Chirality****COMMON EXAMINER TRAP / MISTAKE:**

Overlooking chiral carbons inside saturated ring systems.

MODEL ANSWER / EXAMINER EXPECTATION:

Check carbons in rings: if the two pathways around the ring from that carbon are DIFFERENT, the attached ring paths count as two distinct groups! E.g. 3-methylcyclohexanol has 2 chiral carbons.

FAQ #2: Symmetry Plane in Molecules**Category: Mesocompounds****COMMON EXAMINER TRAP / MISTAKE:**

Assuming any molecule with 2 chiral carbons MUST be optically active.

MODEL ANSWER / EXAMINER EXPECTATION:

If a molecule contains a plane of internal symmetry (meso form), it is achiral and optically inactive despite having 2 chiral centers.

FAQ #3: SN1 vs SN2 Stereochemical Distinction**Category: Mechanism Evidence****COMMON EXAMINER TRAP / MISTAKE:**

Swapping SN1 and SN2 stereochemical products.

MODEL ANSWER / EXAMINER EXPECTATION:

SN1 → planar carbocation → 50:50 attack → RACEMIC MIXTURE (optically inactive). SN2 → backside attack → INVERSION → single optically active enantiomer.

FAQ #4: Physical Properties of Enantiomers**Category: Enantiomer Properties****COMMON EXAMINER TRAP / MISTAKE:**

Claiming enantiomers have different boiling points.

MODEL ANSWER / EXAMINER EXPECTATION:

Enantiomers have IDENTICAL physical properties (b.p., m.p., density, solubility in achiral solvents). They ONLY differ in direction of rotation of plane-polarised light.

FAQ #5: Enzymatic Reactivity of Enantiomers**Category: Biological Action****COMMON EXAMINER TRAP / MISTAKE:**

Expecting both enantiomers to have equal pharmaceutical activity.

MODEL ANSWER / EXAMINER EXPECTATION:

Enzymes and biological receptors are CHIRAL (lock-and-key). Usually only ONE enantiomer fits the active site (e.g. Thalidomide, Ibuprofen).

FAQ #6: Racemate Optical Rotation**Category: Racemic Mixture****COMMON EXAMINER TRAP / MISTAKE:**

Writing that a racemate rotates light by 0.5 degrees.

MODEL ANSWER / EXAMINER EXPECTATION:

A racemic mixture contains EQUAL MOLES of (+) and (-) enantiomers. The opposite rotations cancel out completely → 0.0° optical rotation.

FAQ #7: Drawing 3D Enantiomers**Category: 3D Projection****COMMON EXAMINER TRAP / MISTAKE:**

Drawing mirror images that are superimposable.

MODEL ANSWER / EXAMINER EXPECTATION:

Use wedge (pointing out), dash (pointing in), and two solid lines (in page plane). Ensure the mirror image is non-superimposable upon 180° rotation.

FAQ #8: Electrophilic Addition to Alkenes Stereochemistry**Category: Alkene Addition****COMMON EXAMINER TRAP / MISTAKE:**

Expecting alkene addition to yield a single optically active enantiomer.

MODEL ANSWER / EXAMINER EXPECTATION:

Planar C=C double bond allows attack from top or bottom with equal probability → ALWAYS forms a racemic mixture if a chiral center is created.

FAQ #9: Nucleophilic Addition to Carbonyls Stereochemistry**Category: Carbonyl Addition****COMMON EXAMINER TRAP / MISTAKE:**

Assuming HCN addition to aldehydes gives optically active product.

MODEL ANSWER / EXAMINER EXPECTATION:

Planar C=O carbonyl group is attacked by CN⁻ from top or bottom with equal probability → yields a RACEMIC mixture of hydroxynitrile.

FAQ #10: Polarimeter Source Light**Category: Experimental Setup****COMMON EXAMINER TRAP / MISTAKE:**

Using unpolarised white light in a polarimeter.

MODEL ANSWER / EXAMINER EXPECTATION:

A polarimeter MUST use PLANE-POLARISED light (produced using a Nicol prism / polarising filter). Unpolarised light contains waves vibrating in all planes.